

TECHNICAL PRIMER

Understanding AI-Guided Exploration of a CFD-Informed Combustion-Stability Analog

An Undergraduate-Level Technical Primer

A textbook-style guide to the CFD environment, stability analog, Gaussian-process exploration strategy, matched-budget experiments, scientific follow-up studies, and limitations.

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CFD • Active Learning • Combustion-Stability Analog

Overview

This document explains an autonomous scientific-exploration project in which an AI agent chooses which computational fluid dynamics (CFD) simulations to perform in order to learn the boundary between stable and unstable behavior in a simplified combustion-stability analog.

The most important distinction to understand at the beginning is that the project is not a predictive simulation of a flight rocket engine. OpenFOAM is used to compute physically generated mixing fields in a two-dimensional laminar dual-jet chamber. Quantities extracted from those fields are then passed through a declared Rayleigh-inspired acoustic analog to obtain a scalar stability indicator, σ_{analog} .

The scientific question is therefore not “Can this program predict whether a real rocket engine is stable?” Instead, the question is:

When each physical simulation is expensive and the scientifically interesting regime is relatively rare, can an adaptive AI exploration policy use a fixed simulation budget more effectively than conventional space-filling sampling?

The project compares a Gaussian-process-based active explorer with Latin-hypercube sampling under matched evaluation budgets. It also illustrates a useful pattern for AI-assisted science: autonomous exploration produces an unexpected observation, the observation motivates a controlled follow-up experiment, and sensitivity analysis determines which parts of the finding are robust and which depend on modeling assumptions.

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1 The Big Picture

Suppose a scientist has a computational experiment that takes substantial time or computing resources to evaluate. There may be five, ten, or even hundreds of input parameters. The scientist would like to know where interesting behavior occurs, but evaluating every possible input is impossible.

A conventional strategy is to spread simulation points throughout the input space. Latin-hypercube sampling is one example. It attempts to cover each coordinate reasonably uniformly.

An alternative is *adaptive exploration*. Instead of deciding all simulation points in advance, the algorithm proceeds sequentially:

1. Choose an input condition.
2. Run the expensive simulation.
3. Observe the result.
4. Update a statistical model of the environment.
5. Use the model's prediction and uncertainty to choose the next simulation.
6. Repeat until the simulation budget is exhausted.

This project tests whether the second strategy is advantageous when the scientifically important target is a relatively rare unstable regime.

The overall pipeline is

$$\mathbf{z} \rightarrow \text{OpenFOAM mixing} \rightarrow q_{\text{mix}}(x), \tau, R_{\text{spatial}} \rightarrow \sigma_{\text{analog}}(\mathbf{z}) \rightarrow \text{stable or unstable}$$

where

$$\mathbf{z} = [g, d, a, s, o]$$

is the five-dimensional design vector.

The coordinate x has a completely different meaning: it is physical axial position inside the chamber. Keeping $\mathbf{z}$ and x conceptually separate avoids a great deal of confusion.

2 What Is Actually Being Claimed?

Before discussing the equations, it is important to establish the scientific claim boundary.

2.1 What OpenFOAM provides

OpenFOAM provides a physically generated flow and passive-scalar mixing solution for a two-dimensional dual-jet configuration.

From those computed fields, the project obtains quantities such as

- a mixing-delay estimate τ ,
- a cross-stream scalar-variance profile $q_{\text{mix}}(x)$,
- a spatial overlap R_{spatial} ,
- mixedness and profile compactness.

These quantities genuinely depend on the CFD solution.

2.2 What is constructed afterward

The quantity

$$\sigma_{\text{analog}}$$

is not directly solved by OpenFOAM.

It is computed afterward using a declared acoustic/Rayleigh analog. Therefore, it should be interpreted as a *hypothesis-level stability indicator within the exploration environment*.

It should not be interpreted as

- the growth rate of a real rocket-engine instability,
- an experimentally validated engine stability margin,
- a prediction from three-dimensional reacting LES,
- or a dimensional flight-injector design result.

This distinction is central to the scientific integrity of the project.

3 The Fixed CFD Environment

The exploration is performed in a fixed chamber. The relevant dimensions and reference values used by the implementation are

$$H = 0.020 \text{ m}, \quad L = 0.080 \text{ m}, \quad W = 0.002 \text{ m},$$

where H is chamber height, L is chamber length, and W is the thin extrusion thickness used to represent the nominally two-dimensional domain.

The implementation also specifies

$$\nu = 1.0 \times 10^{-5} \text{ m}^2/\text{s},$$

$$U_{\text{ref}} = 0.18 \text{ m/s},$$

and a mixture-fraction diffusivity of

$$D_Z = 2.5 \times 10^{-5} \text{ m}^2/\text{s}.$$

The chamber geometry and the general physical model remain fixed throughout the search.

What changes from experiment to experiment is the inlet configuration.

4 Background: What Does the CFD Solve?

At an undergraduate level, it is useful to think of the CFD portion as solving two coupled problems.

First, an incompressible fluid velocity field approximately obeys conservation of mass,

$$\nabla \cdot \mathbf{u} = 0,$$

and momentum conservation,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}.$$

Second, a passive scalar representing mixture fraction is transported by the flow. A generic passive-scalar equation is

$$\frac{\partial Z}{\partial t} + \mathbf{u} \cdot \nabla Z = D_Z \nabla^2 Z.$$

Here Z can be interpreted as a marker for which inlet a parcel of fluid came from.

One inlet is assigned approximately

$$Z = 0,$$

while the other is assigned approximately

$$Z = 1.$$

If the two streams remain completely separated, a cross-section will contain strong variations between values near 0 and values near 1.

If they mix thoroughly, the cross-section becomes much more uniform.

This is why variance in Z can be used as a measure of remaining *unmixedness*.

5 The Five-Dimensional Exploration Vector

The AI does not directly modify arbitrary mesh cells. Instead, it chooses five abstract coordinates,

$$\mathbf{z} = [g, d, a, s, o].$$

Each coordinate maps to a physically meaningful change in the inlet configuration.

5.1 Pattern-class analog g

The variable

$$g \in [0, 2]$$

changes the relative locations of the two inlet slots.

Conceptually,

$$g \approx 0$$

resembles a close like-on-like arrangement,

$$g \approx 1$$

resembles a more separated unlike-impinging family, and

$$g \approx 2$$

moves toward a coaxial-like stacked arrangement.

Importantly, these are *analogs*. They are not dimensional injector blueprints.

The slot center positions are interpolated piecewise through representative low, middle, and high pattern configurations.

5.2 Orifice-size-spread analog d

The variable

$$d \in [0, 1]$$

controls relative inlet-slot heights.

The implementation uses

$$h_0 = 0.13H(1 + 0.55d),$$

$$h_1 = 0.13H(1 - 0.55d).$$

When $d = 0$, the slots are equal in size.

As d increases, one grows while the other shrinks.

For $d = 1$, the multiplicative factors are

$$1.55 \quad \text{and} \quad 0.45.$$

Thus, d represents increasing asymmetry in inlet size.

5.3 Impingement analog a

The variable

$$a \in [0, 1]$$

controls the inlet-vector angle.

The implementation defines

$$\theta = (0.12 + 0.70a)(0.70)$$

radians.

This corresponds approximately to an angle range from about

$$7^\circ$$

to

$$47^\circ.$$

The two inlet streams are directed toward one another in the cross-stream direction, so increasing a changes how strongly their trajectories impinge.

5.4 Swirl analog s

The variable

$$s \in [0, 1]$$

is called a swirl analog.

It is essential not to overinterpret this term.

The environment is two-dimensional, so it cannot reproduce true three-dimensional azimuthal swirl. Instead, increasing s changes the relative slot offsets and adds opposing cross-stream velocity.

The implementation defines a spin-like factor

$$\text{spin} = 0.45s.$$

Therefore s should be understood as a controllable two-dimensional stand-in for flow deflection associated with swirl-like behavior.

5.5 Operating analog o

The final coordinate,

$$o \in [0, 1],$$

changes the inlet axial velocities.

The two inlet speeds are

$$u_{0x} = U_{\text{ref}}(0.70 + 0.30o),$$

and

$$u_{1x} = U_{\text{ref}}(1.30 - 0.30o).$$

As o changes, one jet becomes faster while the other becomes slower.

The same variable weakly modifies the acoustic frequency in the later analog, which will be discussed below.

6 How Mixing Is Converted into a Useful Signal

The AI ultimately needs a scalar response telling it whether the current design is stable or unstable.

The first step is to extract useful information from the mixture-fraction field.

6.1 Cross-stream variance

At a fixed axial location x , consider the distribution of Z across the height of the chamber.

The project defines

$$q_{\text{mix}}(x) = \text{Var}_y[Z](x).$$

This quantity is the cross-stream variance of mixture fraction.

If all fluid at that station has nearly the same mixture fraction, then

$$q_{\text{mix}}(x) \rightarrow 0.$$

If strong segregation remains, the variance is larger.

Therefore,

$$q_{\text{mix}}$$

is best described as a *mixing-availability or unmixedness proxy*.

It is not heat release.

This distinction matters because a reacting flame would require chemistry, temperature-dependent kinetics, heat release, species transport, and potentially compressibility. Those processes are not present here.

7 Mixing Delay τ

A useful quantity in combustion-acoustic reasoning is a time delay between injection and some downstream response.

The project constructs an operational mixing delay from the CFD field.

The chamber is divided into

$$N_{\text{bins}} = 24$$

equal-width axial bins.

For every bin, the cross-stream mixture-fraction variance is computed.

The project then searches for the first axial bin whose variance satisfies

$$\text{Var}_y[Z] < V_{\text{crit}},$$

where the default criterion is

$$V_{\text{crit}} = 0.045.$$

Let the center of this first sufficiently mixed bin be x_m .

If no bin reaches the criterion, the final axial bin is used.

A bulk inlet speed is defined as

$$U_b = \frac{1}{2} (|u_{0x}| + |u_{1x}|).$$

The mixing delay is then

$$\tau = \max \left(10^{-4}, \frac{x_m}{U_b} \right).$$

The physical intuition is straightforward.

If mixing occurs farther downstream, x_m is larger and therefore τ is larger.

If the bulk flow becomes faster, the same distance corresponds to a smaller travel time.

8 Spatial Acoustic Overlap

The project declares a simple closed-closed first-longitudinal pressure mode,

$$p(x) = \cos\left(\frac{\pi x}{L}\right).$$

The injector face at $x = 0$ is therefore a pressure antinode because

$$p(0) = 1.$$

The spatial relationship between remaining unmixedness and this acoustic mode is summarized using

$$R_{\text{spatial}} = \frac{\int q_{\text{mix}}(x)p(x) dx}{\int q_{\text{mix}}(x) dx}.$$

This resembles an overlap integral.

If the spatial distribution of mixing availability lines up strongly with the positive part of the pressure mode, R_{spatial} becomes more positive.

If it lies where the pressure mode has the opposite sign, the overlap can be reduced or negative.

Again, this is a declared analog rather than a solved acoustic eigenproblem.

9 Additional Mixing Quantities

The project also calculates compactness,

$$C = \frac{\max_x q_{\text{mix}}(x)}{\text{mean}_x q_{\text{mix}}(x)},$$

which measures how concentrated the unmixedness profile is.

A mixedness quantity U_m is evaluated near the 45% axial station.

These quantities affect the interaction index used by the acoustic analog.

10 The Rayleigh-Inspired Stability Analog

The CFD quantities are now converted into one scalar stability indicator.

The interaction index is

$$n = 0.50 + 0.28 \tanh(C - 1) + 0.16(1 - U_m) + 0.10(o - 0.5)^2.$$

Several details are worth noticing.

First, g , d , a , and s do not appear directly in this formula.

They influence n only indirectly by changing the OpenFOAM mixing field and therefore quantities such as C and U_m .

This is useful because the environment does not simply say,

“If g is large, make the point unstable.”

Instead, the geometry changes the CFD solution, and the CFD solution changes the features supplied to the analog.

The angular frequency is

$$\omega = \omega_0(0.92 + 0.16o),$$

with

$$\omega_0 = 11.$$

The stability indicator is then

$$\sigma_{\text{analog}} = \alpha n R_{\text{spatial}} \cos(\omega\tau) - D$$

with

$$\alpha = 1.45$$

and

$$D = 0.08.$$

11 Why Does the Cosine Matter?

The term

$$\cos(\omega\tau)$$

encodes phase.

Imagine a periodic acoustic oscillation.

If the mixing or heat-release-like response arrives at a phase that reinforces the acoustic pressure oscillation, the coupling can be positive.

If it arrives out of phase, it can oppose the oscillation.

The exact real-engine physics is substantially more complicated, but this simple phase relationship is what allows a change in CFD-derived mixing delay to move the analog between stable and unstable behavior.

This is also one reason a monotonic change in geometry need not produce a monotonic change in stability.

If a geometry parameter changes τ , then

$$\omega\tau$$

changes, and a cosine is inherently nonmonotonic.

Thus, it is entirely possible for a parameter sweep to produce

$$\text{unstable} \rightarrow \text{stable} \rightarrow \text{unstable}.$$

12 Stable Versus Unstable

The AI works directly with

$$\sigma_{\text{analog}}.$$

The stability boundary is

$$\sigma_{\text{analog}} = 0.$$

The convention is

$$\sigma_{\text{analog}} > 0 \quad \Rightarrow \quad \text{unstable},$$

and

$$\sigma_{\text{analog}} \leq 0 \quad \Rightarrow \quad \text{stable}.$$

The implementation also defines an amplitude-like transformed quantity S .

For positive σ ,

$$S = \exp(3.2\sigma),$$

while for negative σ ,

$$S = \frac{1}{1 - 4\sigma}.$$

At

$$\sigma = 0,$$

both definitions give

$$S = 1.$$

Therefore the equivalent threshold is

$$S = 1.$$

However, the exploration algorithm fits and searches σ directly, because $\sigma = 0$ is the natural level-set boundary.

13 What Is a Level Set?

Suppose a scalar function is

$$f(\mathbf{z}).$$

A level set is the set of all points for which

$$f(\mathbf{z}) = c$$

for some constant c .

Here,

$$f(\mathbf{z}) = \sigma_{\text{analog}}(\mathbf{z}),$$

and the target constant is zero.

Therefore the desired boundary is

$$\mathcal{B} = \{\mathbf{z} : \sigma_{\text{analog}}(\mathbf{z}) = 0\}.$$

The AI is therefore not merely trying to locate one global optimum.

It is trying to learn the shape of a boundary separating two regimes.

14 Why Use a Gaussian Process?

Each CFD evaluation is expensive relative to an algebraic function evaluation.

The algorithm therefore maintains a statistical surrogate.

A Gaussian process gives, at a candidate point $\mathbf{z}$,

$$\mu(\mathbf{z}),$$

the predicted mean value of σ , and

$$s(\mathbf{z}),$$

the predictive standard deviation.

The mean tells us what the model currently expects.

The standard deviation tells us where the model is uncertain.

This combination is useful for deciding where another expensive experiment could provide the most information.

The implementation uses a Matérn kernel with smoothness

$$\nu = 2.5,$$

together with constant and white-noise components.

For understanding the project, the exact kernel is less important than the idea that nearby design points are assumed to have correlated responses while the model retains uncertainty where evidence is sparse.

15 The Initial Experimental Design

The AI is not given a specially favorable initial set.

Its initial points are generated using Latin-hypercube sampling.

In the primary live-CFD comparison, five initial points are used.

Latin-hypercube sampling divides each parameter dimension into strata and tries to ensure broad coverage.

This means the comparison is not

carefully designed AI start versus bad random baseline.

Instead, both approaches begin from a reasonable space-filling philosophy.

The difference appears after initialization.

16 How the AI Chooses the Next Query

The algorithm operates in two conceptual modes.

16.1 Mode 1: find the missing regime

Suppose every observed point so far is stable.

Then the most urgent scientific question is not

“Where exactly is the stability boundary?”

because the algorithm does not yet know whether an unstable regime exists.

Instead, it searches for a point that could plausibly have a large positive σ .

The implementation uses an upper-confidence-type score,

$$A_{\text{unstable}} = \mu + 1.9s.$$

A point can receive a high score because

- the model predicts positive σ , or
- the model is highly uncertain and therefore could hide an unstable region.

The symmetric idea is used if only unstable points have been observed.

16.2 Mode 2: reconstruct the boundary

Once both stable and unstable regimes have been observed, the algorithm uses a straddle-style acquisition,

$$A_{\text{straddle}} = 1.96s - |\mu|.$$

This score becomes large when

1. the model is uncertain, so s is large, and
2. the predicted mean is near the threshold, so $|\mu|$ is small.

Those are exactly the points likely to be informative about

$$\sigma = 0.$$

A small probability of random exploration is retained so that the algorithm does not become completely trapped by its current surrogate.

17 The Baseline: Latin-Hypercube Sampling

A scientific comparison requires a reference frame.

The main baseline is Latin-hypercube sampling.

For the live OpenFOAM study,

$$B_{\text{AI}} = B_{\text{LHS}} = 16.$$

That means each method receives exactly 16 expensive evaluations.

The same

- parameter bounds,
- OpenFOAM environment,
- stability analog,
- solver iteration policy,
- and independent test set

are used.

Therefore a performance difference is intended to measure the value of adaptive selection rather than extra computational budget.

18 Atlas Experiments Versus Live OpenFOAM Experiments

The repository contains two related but different kinds of exploration.

18.1 The OpenFOAM mixing atlas

A set of OpenFOAM simulations is precomputed and used to train interpolators for mixing quantities.

This allows the exploration environment to respond quickly.

Interactive dashboard experiments use this atlas so a judge can run a campaign without waiting for many CFD simulations.

18.2 Fresh live-CFD campaigns

The main confirmation experiment does not depend only on the interpolator.

For these campaigns, each selected design is passed to a fresh OpenFOAM simulation.

This is why the live-CFD results are particularly important.

The fast atlas shows whether the concept works repeatedly and interactively.

The live runs show that the performance advantage survives when each evaluation is a fresh solver call.

19 Independent Hold-Out Data

A model could appear successful merely because it performs well on the same data used to construct it.

To avoid that problem, campaign models are scored on an independent OpenFOAM test set.

The project uses a 24-case hold-out set stored in

`artifacts/of_test.json`

This set is used for scoring rather than selecting the next campaign point.

That separation is analogous to training and test data in machine learning.

20 Why Overall Accuracy Is Not Enough

Suppose 80% of an environment is stable and only 20% is unstable.

A useless classifier could predict

stable everywhere

and achieve

80%

overall accuracy.

But it would find none of the dangerous unstable cases.

That is why the primary metric is unstable recall.

If

TP_U

denotes correctly identified unstable cases and

FN_U

denotes unstable cases incorrectly predicted as stable, then

$$\text{Recall}_U = \frac{TP_U}{TP_U + FN_U}$$

measures the fraction of true unstable cases recovered.

21 Precision and F1

Recall by itself could also be abused.

A model could simply label nearly everything unstable.

It might then have high recall but many false alarms.

Therefore unstable precision is also reported,

$$\text{Precision}_U = \frac{TP_U}{TP_U + FP_U}.$$

The F1 score combines precision and recall,

$$F1_U = 2 \frac{\text{Precision}_U \text{Recall}_U}{\text{Precision}_U + \text{Recall}_U}.$$

The purpose of reporting all three is to distinguish

“The AI genuinely finds the rare unstable regime”

from

“The AI simply predicts unstable everywhere.”

22 Near-Boundary Error

Another metric examines held-out cases already known to lie close to the stability threshold.

Define

$$\mathcal{B}_{0.20} = \{i : |\sigma_i| < 0.20\}.$$

The reported near-boundary error is

$$E_{\sigma, \text{boundary}} = \frac{1}{|\mathcal{B}_{0.20}|} \sum_{i \in \mathcal{B}_{0.20}} |\hat{\sigma}(\mathbf{z}_i) - \sigma(\mathbf{z}_i)|.$$

This is an error in predicted growth-rate analog.

It is not a geometric Hausdorff distance between two surfaces.

23 Results: Atlas Study

The fast atlas experiment uses 24 seeds and a budget of 16 evaluations.

The mean results are

Metric	AI	LHS	Random
Unstable recall	0.59	0.46	0.38
Unstable precision	0.88	0.91	0.79
Unstable F1	0.69	0.60	0.47
Unstable evaluations found	7.5	4.46	3.54
Near-boundary σ MAE	0.123	0.167	0.195
Volume accuracy	0.825	0.788	0.753

Several conclusions follow.

First, adaptive exploration finds substantially more unstable observations.

Second, it improves unstable recall.

Third, precision does not improve; in fact LHS has slightly higher mean precision in this atlas study.

That is scientifically useful because it narrows the correct claim.

The claim is not

“AI dominates every metric.”

The claim is

“Adaptive exploration is advantageous for rare-regime recovery under a limited evaluation budget.”

24 Results: Eight Live OpenFOAM Campaigns

The stronger confirmation consists of eight matched live OpenFOAM campaigns.

The seeds are

8, 11, 14, 19, 23, 26, 32, 35.

Each method receives 16 evaluations per seed.

The aggregate results are approximately

Metric	AI	LHS
Unstable recall	0.68 ± 0.13	0.33 ± 0.21
Unstable precision	0.86 ± 0.11	0.85 ± 0.35
Unstable F1	0.76 ± 0.12	0.46 ± 0.25
Unstable evaluations found	7.5 ± 1.2	5.9 ± 1.6
Volume accuracy	0.84 ± 0.07	0.75 ± 0.08
Near-boundary σ MAE	0.134 ± 0.031	0.134 ± 0.046

AI has higher unstable recall in

7/8

seeds.

It also samples more unstable evaluations in

7/8

seeds.

The F1 result follows the same 7-of-8 pattern.

However, mean unstable precision is essentially tied.

The near-boundary growth-rate MAE is also essentially tied in the mean.

Seed 35 reverses the primary result and is retained rather than removed.

This is important scientifically.

A credible conclusion should acknowledge failure cases.

25

The Most Interesting Scientific Observation: Seed 14

The method comparison is only one part of an Open Exploration project.

The more interesting scientific story begins with seed 14.

During full five-dimensional adaptive exploration, the AI encountered unstable conditions at both relatively low and relatively high values of the pattern-class variable g .

Examples from the live exploration include approximately

t	g	d	a	s	o	σ
2	0.10	0.99	0.07	0.45	0.12	+0.32
3	1.80	0.35	0.87	0.21	0.74	+0.06
6	0.52	0.51	0.92	0.27	0.90	+0.21
14	1.92	0.93	0.73	0.04	0.38	+0.40

These points do not prove that two disconnected five-dimensional unstable regions exist.

The other coordinates differ.

However, they produce a legitimate scientific question:

Could stability be nonmonotonic with respect to pattern class g ?

26 From Autonomous Observation to Controlled Experiment

The correct response to an unexpected multivariable observation is not to immediately declare causality.

Instead, the scientist isolates the suspected variable.

Therefore a controlled OpenFOAM g -sweep was performed while the other conditions were held fixed.

The resulting one-dimensional slice showed the qualitative sequence

unstable $\rightarrow$ stable $\rightarrow$ unstable

as g increased.

Thus, on that declared low-swirl slice, the stability analog is *nonmonotonic* and contains two separated unstable intervals.

This is a stronger statement than the original five-dimensional observation because the controlled sweep isolates g .

But it is still a one-dimensional statement.

It does not establish disconnected topology throughout the full five-dimensional domain.

27 The Swirl Follow-Up

A separate controlled sweep varies the swirl analog s while remaining within a like-on-like family.

Over the tested range, increasing s reduces σ_{analog} .

However, it does not cross the stability boundary.

Therefore the supported statement is approximately

Increasing the swirl analog reduced the instability indicator on the tested like-on-like slice.

It is not justified to say

More swirl always makes every configuration stable.

28 Why Sparse Five-Dimensional Comparisons Are Not Causal Proof

This point is important for scientific reasoning.

Suppose one observed point has

$$s = 0.2$$

and another has

$$s = 0.8.$$

If the second is more stable, one might be tempted to conclude that increasing s caused the increased stability.

But if g , d , a , and o also changed, the comparison is confounded.

Therefore sparse multivariable observations should be treated as

hypothesis-generating evidence,

not definitive causal evidence.

Controlled slices are what allow stronger one-variable statements.

29 Sensitivity Analysis

A useful scientific discovery should be tested against reasonable changes in assumptions.

The project performs two classes of sensitivity analysis.

29.1 Acoustic-analog sensitivity

The CFD mixing fields are kept fixed while the assumed damping D and base frequency ω_0 are changed.

The tested damping perturbation is approximately

$$D \in \{0.8D_0, D_0, 1.2D_0\}.$$

The tested frequency perturbation is

$$\omega_0 \in \{0.9\omega_{0,\text{base}}, \omega_{0,\text{base}}, 1.1\omega_{0,\text{base}}\}.$$

The two-interval g -slice structure survives the damping perturbations.

At a 10% lower frequency, two intervals also remain.

At a 10% higher frequency, however, the high- g unstable interval disappears.

This is an important negative result.

It means that the second interval is not universal within the analog.

30 Mixing-Delay-Definition Sensitivity

The definition of τ is itself a modeling choice.

Therefore the project perturbs

$$V_{\text{crit}} \in \{0.035, 0.045, 0.055\}$$

and

$$N_{\text{bins}} \in \{16, 24, 32\}.$$

Changing the number of axial stations in this tested range does not eliminate the two-interval structure.

Using

$$V_{\text{crit}} = 0.055$$

also preserves it.

But tightening the criterion to

$$V_{\text{crit}} = 0.035$$

removes the high- g unstable interval.

Again, the appropriate conclusion is conditional:

The high- g branch exists under the declared baseline definition but is sensitive to a stricter mixing criterion.

31 What the Final Scientific Conclusion Should Be

The project supports three distinct conclusions.

31.1 Methodological conclusion

Under the same small simulation budget, adaptive experiment selection recovers the rare unstable regime more effectively than Latin-hypercube sampling in this environment.

The strongest evidence is the eight matched fresh-OpenFOAM campaigns, in which AI has higher unstable recall in seven of eight seeds.

31.2 Exploration conclusion

Autonomous exploration can do more than improve an aggregate machine-learning metric.

In seed 14, it produced unexpected unstable observations at both low and high values of g .

Those observations suggested a specific follow-up scientific question.

31.3 Scientific-structure conclusion

A controlled low-swirl OpenFOAM sweep subsequently established a nonmonotonic dependence on g with two separated unstable intervals on that one-dimensional slice.

Sensitivity tests show that the high- g interval is conditional on some modeling assumptions.

Therefore it should be viewed as a candidate piece of analog structure rather than a universal combustion-physics law.

32 Where the Actual Data Live

The actual experiment history is stored in JSON Lines files.

For the primary seed-14 AI live-OpenFOAM campaign:

`artifacts/cfd_study_s14/ai/exploration.jsonl`

The matched Latin-hypercube experiment is

`artifacts/cfd_study_s14/baseline/exploration.jsonl`

The aggregate eight-seed summary is

`artifacts/cfd_live_summary.json`

The independent test set is

`artifacts/of_test.json`

The controlled pattern-class sweep is

`artifacts/g_sweep.json`

The controlled swirl sweep is

`artifacts/swirl_sweep.json`

The acoustic sensitivity results are

`artifacts/sensitivity.json`

and the mixing-delay-definition sensitivity results are

`artifacts/tau_sensitivity.json`

33 How to Read One Exploration-Log Row

Each line in an `exploration.jsonl` file corresponds to one simulation chosen by that campaign.

Conceptually, a row contains fields such as

```
{
  "g": ...,
  "d": ...,
  "a": ...,
  "s": ...,
  "o": ...,
  "tau": ...,
  "R_spatial": ...,
  "sigma": ...,
  "stable": ...,
  "backend": "openfoam",
  "t": ...,
  "acquisition": ...
}
```

The first five numbers answer

What experiment did the agent choose?

The CFD-derived quantities answer

What physical mixing behavior did that experiment produce?

The σ and stability fields answer

How did the declared stability analog classify it?

The time step t answers

When in the sequential exploration did this observation occur?

The acquisition value provides information about why an adaptive query was preferred by the exploration policy.

34 Campaign Metadata Versus Raw Data

A file such as

`campaign.json`

should not be confused with the raw exploration history.

The JSONL file contains the individual evaluated points.

The campaign metadata file summarizes items such as

- method,
- random seed,
- evaluation budget,
- number of valid evaluations,
- hypotheses,
- interpreted discoveries.

Thus, if a reviewer asks for the primary evidence, the best file to show is the JSONL exploration log.

If a reviewer asks for the interpretation of the campaign, show `campaign.json`.

35 Why This Is an Open-Exploration Project

A conventional supervised-learning project might proceed as follows:

large fixed dataset → train model → report test accuracy.

This project is structurally different.

The algorithm participates in determining which scientific observations are generated:

current evidence → choose next experiment → run simulator → update evidence → choose again.

That feedback loop is why the system is naturally described as an *exploration environment*.

The output is not only a fitted predictor.

It is also an experiment trajectory and a sequence of hypotheses and follow-up questions.

36 A Simple Analogy

Imagine that a mountain range is hidden by fog.

Red regions are dangerous terrain.

A traditional survey strategy places measuring stations evenly across the map.

An adaptive strategy begins with some evenly spread measurements but then notices clues that dangerous terrain may lie in a certain direction.

It spends more of its limited measurements there.

The objective is not necessarily to produce the best average elevation estimate everywhere.
 It may instead be to identify dangerous regions before the measurement budget runs out.
 In this project,

$$\text{mountain terrain} \longleftrightarrow \sigma_{\text{analog}}(\mathbf{z}),$$

$$\text{measurement} \longleftrightarrow \text{OpenFOAM evaluation},$$

and

$$\text{dangerous terrain} \longleftrightarrow \sigma_{\text{analog}} > 0.$$

This analogy explains why rare-regime recall is more important than only global accuracy.

37 Limitations

A careful interpretation should retain the following limitations.

1. The CFD environment is two-dimensional.
2. The flow model is laminar and non-reacting.
3. Mixture-fraction variance is unmixedness, not heat release.
4. The pressure mode is declared rather than obtained from a full combustor acoustic eigenproblem.
5. The acoustic constants are analog parameters rather than measured engine quantities.
6. The fast atlas is an interpolation of a finite set of OpenFOAM simulations.
7. The second unstable interval is established on a controlled one-dimensional slice, not as a disconnected five-dimensional component.
8. The high- g interval disappears for some plausible perturbations of the analog assumptions.

These limitations do not invalidate the exploration experiment.

They define exactly what the experiment establishes.

38 Natural Next Research Steps

Several follow-up studies would make the environment progressively more physical.

One could replace the passive-scalar mixing proxy with reacting chemistry.

One could solve an acoustic eigenproblem rather than assuming the closed-closed longitudinal mode.

One could perform mesh and solver-convergence studies concentrated around the candidate boundary.

One could extend the geometry to three dimensions.

One could compare several adaptive scientific-exploration algorithms under the exact same CFD budget.

The present environment is therefore useful not only as a single result but as a controlled benchmark for studying how AI should allocate expensive scientific experiments.

39 Final Summary

The entire project can be understood as five linked ideas.

First, a five-dimensional design vector changes inlet geometry and velocity in a fixed OpenFOAM mixing environment.

Second, OpenFOAM produces physically generated mixing fields.

Third, mixing delay and spatial overlap are converted through a declared Rayleigh-inspired model into a stability indicator,

$$\sigma_{\text{analog}}.$$

Fourth, a Gaussian-process agent chooses new experiments adaptively in order to find the rare unstable regime and reconstruct the

$$\sigma_{\text{analog}} = 0$$

boundary.

Fifth, the resulting experiment history is compared with an equal-budget Latin-hypercube reference frame.

The strongest methodological observation is that adaptive exploration has higher unstable recall in seven of eight matched live OpenFOAM campaigns.

The strongest scientific exploration story is that an unexpected seed-14 observation motivated a controlled pattern-class sweep, which revealed a nonmonotonic response with two separated unstable intervals on a specific low-swirl slice.

The strongest scientific caveat is that the high- g interval is model-dependent and disappears under some sensitivity perturbations.

Therefore the most defensible one-sentence conclusion is:

Within the declared CFD-informed stability analog, adaptive experiment selection uses a limited simulation budget more effectively to recover a rare unstable regime and can convert unexpected multivariable observations into targeted scientific follow-up experiments.